

Curriculum Vitae

Yeon-Jae Kim, M.S.

Research Associate

Department of Mechanical Engineering, Engineering Research Institute

Yonsei University, Seoul, Republic of Korea

E-mail: yeonjaekim@yonsei.ac.kr

Research Interests

■ 3D-Aware Robot Learning for Real-World Execution

- Learning-based robotic assembly and manipulation using 3D object understanding
- Generation of smooth, executable trajectories satisfying joint-space and end-effector constraints
- Online collision-aware trajectory generation using real-time 3D scene representations

■ Generative Models for 3D Design and Manufacturing

- Diffusion-guided 3D part assembly, spatial reasoning, and pose optimization
- Intelligent design-to-manufacturing automation for adaptive and flexible production

Education

09/2024 – 08/2026	M.S. Mechanical Engineering Yonsei University, Republic of Korea <i>Thesis:</i> Generative Framework for Automated Conceptual Design in Multi-variant Product Manufacturing. <i>Advisor:</i> Prof. Byung-Kwon Min
03/2022 – 08/2024	B.S. Mechanical Engineering Yonsei University, Republic of Korea
03/2019 – 02/2021	B.S. Mechanical Engineering Dankook University, Republic of Korea

Research Experience

09/2026 – Present	Research Associate Manufacturing & Mechatronics Lab, Engineering Research Institute, Yonsei University <i>Advisor:</i> Prof. Byung-Kwon Min ▪ Online collision-aware trajectory generation using real-time 3D scene representations
09/2024 – 08/2026	Graduate Research Assistant Manufacturing & Mechatronics Lab, Yonsei University <i>Advisor:</i> Prof. Byung-Kwon Min <i>Project:</i> Development of a Hyperconnected Manufacturing Platform Based on Generative Manufacturing Intelligence

Funded by the Ministry of Trade, Industry and Resources (MOTIR), Republic of Korea

- Developed a generative AI-based framework for automated conceptual design
- Designed and implemented methods for automated 3D part assembly
- Applied diffusion-based generative models to optimize 3D design parameters

03/2023 - 02/2024 Undergraduate Research Assistant

Human-Centered AI Robotics Lab, Yonsei University

Advisor: Prof. Dongjun Shin

- Designed and experimentally evaluated a continuous variable transmission mechanism for twisted string actuators

Teaching Experience

09/2024 – 12/2024 Teaching Assistant (Mechatronics)

Department of Mechanical Engineering, Yonsei University

- Lectures on basic embedded systems, motor control, electrical circuits
- Led laboratory sessions and provided Q&A support to students

Awards and Honors

1. KSPE 2026 Spring Outstanding Presentation Award

Korean Society for Precision Engineering

Journal Publications

1. Yeon-Jae Kim, Sangwon Kang, Byung-Kwon Min, "VA-Assembly: View-Aligned Score Guidance for Training-Free 3D Part Assembly." *Submitted* to International Journal of Computer Vision, June 2026.
2. In-Wook Oh, Sangwon Kang, Hee-Min Shin, Yeon-Jae Kim, Soon-Hong Hwang, Byung-Kwon Min, "A Graph Transformer-based agent for feature-free machining process planning." *Submitted* to Journal of Manufacturing Systems, September 2026.

Conferences and Talks

1. Yeon-Jae Kim, Sangwon Kang, Byung-Kwon Min, "6-DOF Robot Motion Planning Using an Adaptive Gaussian Collision Model." Poster presentation, Korean Society of Precision Engineering (KSPE) Fall Conference, December 2026. (Accepted)
2. Yeon-Jae Kim, Sangwon Kang, Byung-Kwon Min, "3D Part Arrangement for Automated Conceptual Design in Multi-variant Product Manufacturing." Oral presentation, Korean Society of Precision Engineering (KSPE) Spring Conference, May 2026.
 - Oral presentation in a special session, "Autonomous Manufacturing"
3. In-Wook Oh, Sangwon Kang, Hee-Min Shin, Yeon-Jae Kim, Byung-Kwon Min, "3D CAD Model-Based Process Planning Using a Graph Transformer" Oral presentation, Korean Society of Precision Engineering (KSPE) Spring Conference, May 2026.
4. Sangwon Kang, In-Wook Oh, Yeon-Jae Kim, Byung-Kwon Min, "Geometry-Aware Trajectory Generation

on NURBS Curves under Machine Kinematic Constraints” Poster presentation, Korean Society of Precision Engineering (KSPE) Spring Conference, May 2026.

5. In-Wook Oh, Sangwon Kang, Hee-Min Shin, Yeon-Jae Kim, Byoung-Seok Kim, Soon-Hong Hwang, Byung-Kwon Min, “Machining Process Planning Model based on Graph Transformer.” Poster presentation, International Conference on Precision Engineering and Sustainable Manufacturing (PRESM), July 2025.

Patents

● Pending

1. Byung-Kwon Min, Yeon-Jae Kim, Sangwon Kang, In-Wook Oh, Soon-Hong Hwang, Hee-Min Shin, and Byoung-Seok Kim. “GENERATIVE AI-BASED PRODUCT DESIGN AUTOMATION AND THREE DIMENSIONAL PART PLACEMENT OPTIMIZATION METHOD.” PCT International Patent Application No. PCT/KR2025/018683, filed Nov 13, 2025.
2. Byung-Kwon Min, Sangwon Kang, In-Wook Oh, Soon-Hong Hwang, Hee-Min Shin, Byoung-Seok Kim, and Yeon-Jae Kim. “SYSTEM AND METHOD FOR AUTOMATICALLY GENERATING MACHINING PROCESS PLANS AND MANUFACTURING CODES.” PCT International Patent Application No. PCT/KR2025/018682, filed Nov 13, 2025.
3. Byung-Kwon Min, Yeon-Jae Kim, Sangwon Kang, In-Wook Oh, Soon-Hong Hwang, Hee-Min Shin, and Byoung-Seok Kim. “GENERATIVE AI-BASED PRODUCT DESIGN AUTOMATION AND THREE-DIMENSIONAL PART PLACEMENT OPTIMIZATION METHOD.” Korean Patent Application No. 10-2025-0127418, filed Sep 8, 2025.
4. Byung-Kwon Min, Sangwon Kang, In-Wook Oh, Soon-Hong Hwang, Hee-Min Shin, Byoung-Seok Kim, and Yeon-Jae Kim. “DEVICE AND METHOD FOR GENERATING MANUFACTURING PROCESS PLANNING DATA SET USING SOFTWARE.” Korean Patent Application No. 10-2025-0073582, filed Jun 5, 2025.
5. Byung-Kwon Min, Sangwon Kang, In-Wook Oh, Soon-Hong Hwang, Hee-Min Shin, Byoung-Seok Kim, and Yeon-Jae Kim. “SYSTEM AND METHOD FOR AUTOMATICALLY GENERATING MACHINING PROCESS PLANS AND MANUFACTURING CODES.” Korean Patent Application No. 10-2025-0073582, filed Jun 5, 2025.

Technical Skills

- Programming: Python, MATLAB, Simulink, C/C++
- AI/ML: PyTorch, Diffusion models, Gaussian splatting
- Design: SolidWorks, Blender
- Hardware: Jetson, Raspberry Pi, Arduino

Languages

- English: Fluent
- Korean: Native